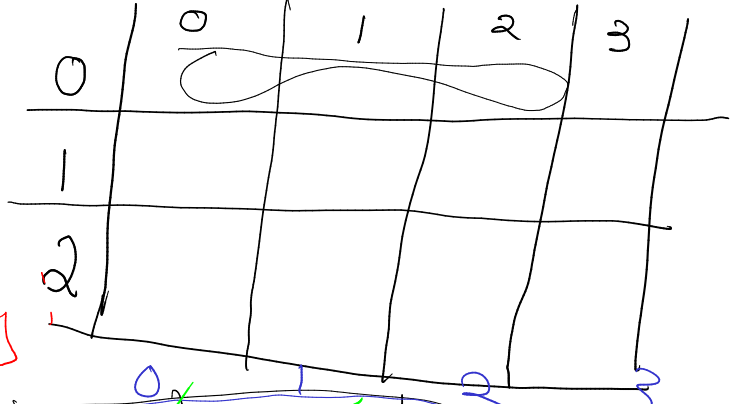
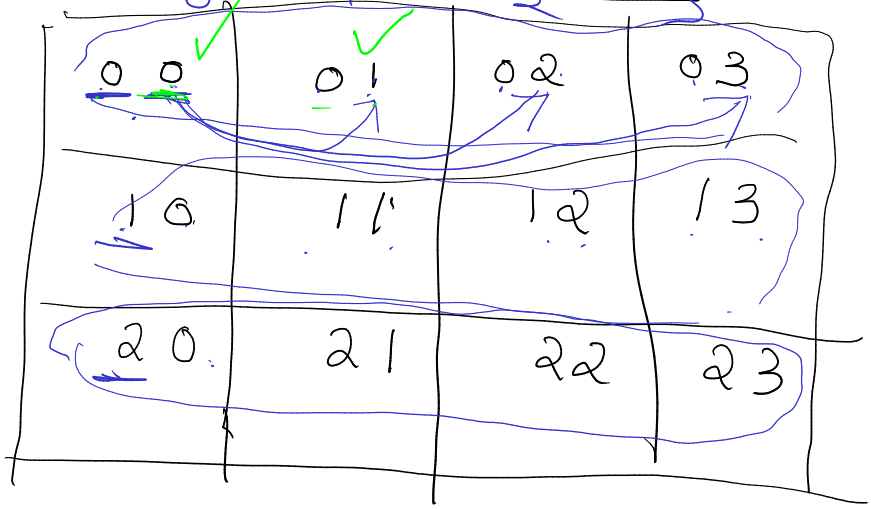
arr[3][4]



1) WAP to Sort all the elements (row wise) in ascending order




```
for(i=0; i<row; i++)
{
    for(j=0; j<col; j++)
    {
        for(k=j+1; k<col; k++)
        {
            if(arr[i][j] > arr[i][k])
            {
                temp = a[i][j];
                a[i][j] = a[i][k];
                a[i][k] = temp;
            }
        }
    }
}
```

i=0

j=0 9 9

arr[0][0] > arr[0][1]

arr[0][0] > arr[0][2]

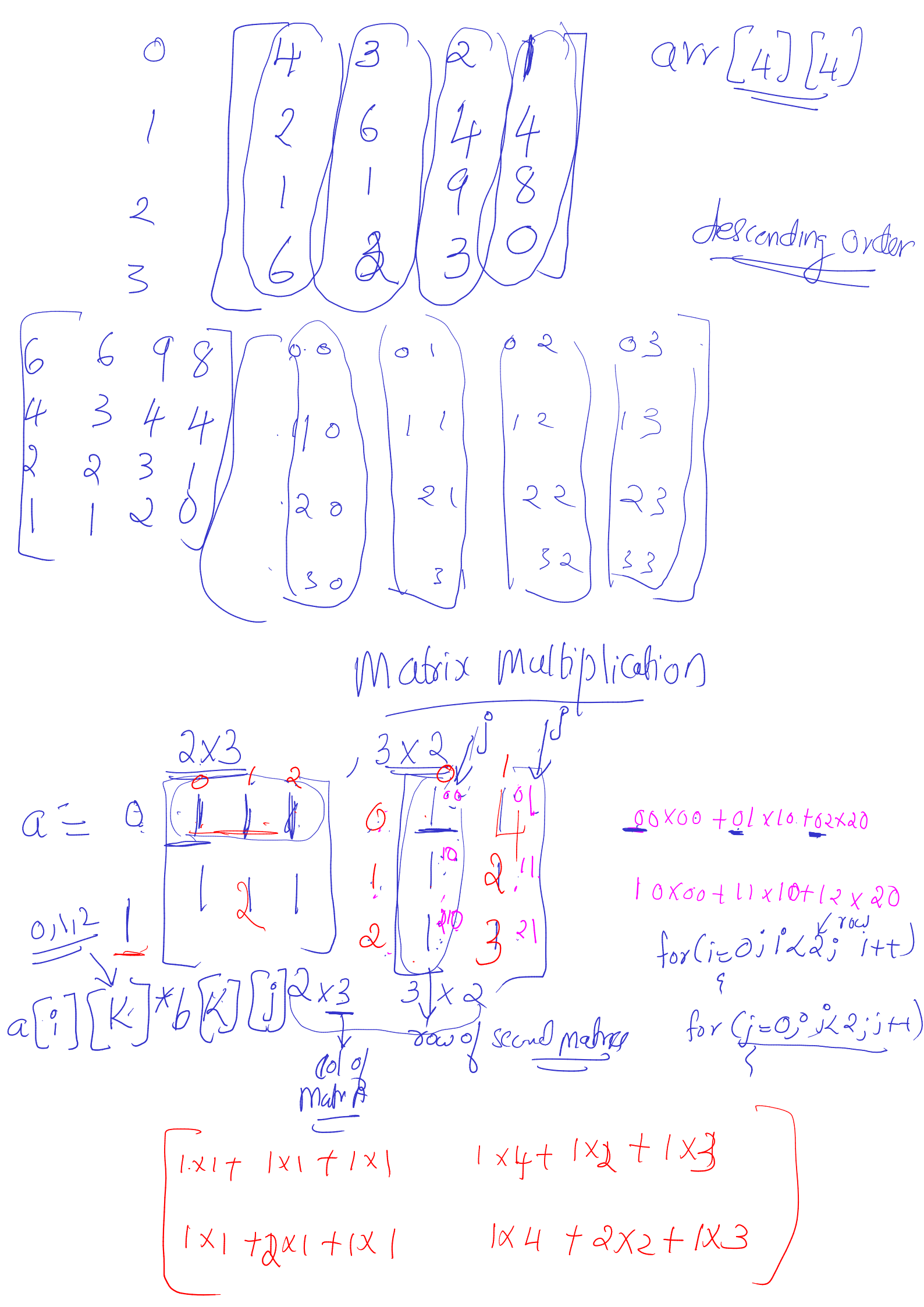
arr[0][0] > arr[0][3]

j=1

arr[0][1] > arr[0][2]

arr[0][1] > arr[0][3]

2). WAP to Sort elements (column wise) in descending order.


arr[4][4]
descending order
Matrix multiplication
2x3
3x2
a = a
0,1,2
1
a[i][k]*b[k][j]
2x3
3x2
col of matrix
row of second matrix
00x00 + 01x10 + 02x20
10x00 + 11x10 + 12x20
for(i=0;i<2;i++) row
for(j=0;j<2;j++)
1x1 + 1x1 + 1x1
1x4 + 1x2 + 1x3
1x1 + 2x1 + 1x1
1x4 + 2x2 + 1x3

```
for(i=0; i<2; i++)
  {
    for(j=0; j<2; j++)
      {
        sum=0;
        for(k=0; k<3; k++)
          {
            sum = sum + a[i][k]*b[k][j];
          }
        c[i][j]=sum;
      }
  }
```

00 01 02

$$\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

a[0][0] x b[0][0]
+

i=0, i<2 ✓ run
j=0, j<2 → col a=
K=0, K<3 // 3 time loop will run
sum=0

$$\begin{bmatrix} 0 & 1 & 2 \\ 00 & 01 & 02 \\ 1 & 1 & 1 \\ 10 & 11 & 12 \\ 1 & 2 & 1 \end{bmatrix}_{2} \quad b: \begin{bmatrix} 0 & 1 \\ 00 & 01 \\ 1 & 3 \\ 10 & 11 \\ 2 & 4 \\ 2^{a}1 & 5 \end{bmatrix}$$

Sum = Sum + a[i][k] * b[k][j] j=0 Crd=0

Sum = Sum + a[0][0] * b[0][0]

Sum = 0 + 1 * 1

Sum = 1

K=1

Sum = Sum + a[i][k] * b[k][j]

Sum = Sum + a[0][1] * b[1][0]

Sum = 1 + 1 * 2

Sum = 1 + 2 => 3

K-loop → 3 times

K=2 , 2 < 3

Sum = Sum + a[i][k] * b[k][j]

Sum = Sum + a[0][2] * b[2][0]

Sum = 3 + 1 * 1

Sum = 4

c[i][j] = sum;

i=0, j=0

k=0, 1, 2

c[i][j] => c[0][0]

c[2][2]

c[0] [00 01 / 4 / 10 11]

Sparse Matrix => Sparse(0) => [3]

5[10][3]

	0	1	2
	0	1	2
5	row	Col	Value
0	0	0	1
1	0	2	2
2	1	3	3
3	2	0	4
4	2	2	5
5	3	0	2

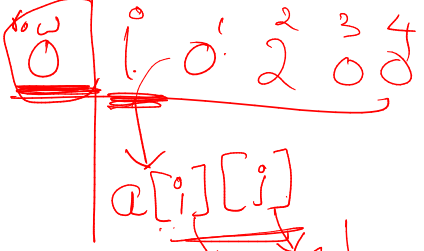
Fixed

	0	1	2	3	4
0	1	0	2	0	0
1	0	0	0	3	0
2	4	0	5	0	0
3	2	0	0	4	0

k=0

```
for(i=0; i<4; i++)
{
    for(j=0; j<5; j++)
    {
        if(          )
        {
        =
        =
        }
    }
}
```

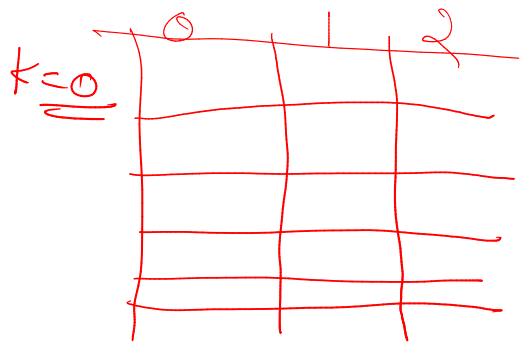
i = row
j = col



// check for non-zero element

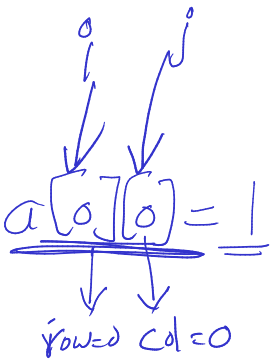
s[10][3]

s[0][0] ⇐ row
s[0][1] ⇐ col
s[0][2] ⇐ value



a[0][0] a[0][1] a[0][2] a[0][3]
4j 4j 4j 4j

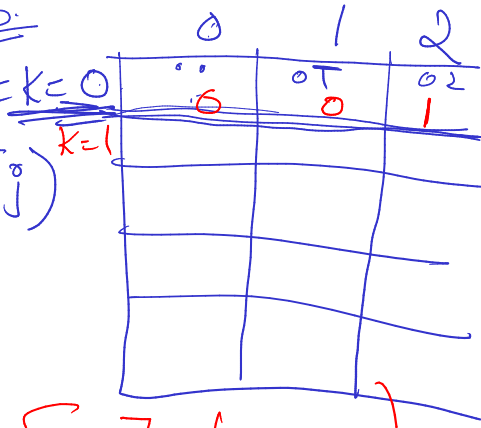
i=
j= 0, 1, 2, 3 ⇒ 4



k=0 i=0 j=0

```
if(a[i][j] != 0)
{
    s[k][0] = i; (row)
    s[k][1] = j
    s[k][2] = a[i][j]
    k++;
}
```

s:



s[0][3]

s[0][0] ⇒ 0th row of col
s[0][1] ⇒ 0th row of 0
s[0][2] ⇒ 0th row and 0 col

j=1

```
if(a[i][1] != 0)
{
```